MIL-C-19256F
26 October 1984
SUPERSEDING
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30 November 1977

MILITARY SPECIFICATION

CLOTH, TWILL, NYLON (LOW COUNT, 3.5 OUNCE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of nylon cloth.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

SPECIFICATIONS

FEDERAL

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Application for copies should be addressed to the Federal Trade Commission, Washington, DC 25380.)

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material. (See 6.5).

3.3.1 Nylon yarn The yarn for both the warp and filling shall be first quality, continuous filament, 210 denier, bright luster and shall have a minimum of 30 filaments when tested as specified in 4.5.

3.4 Physical requirements The finished cloth shall conform to the physical requirements of table I when tested as specified in 4.5.

Table I - Physical Requirements

Weight, oz/yd^2 (g/m^2)		Yarns/inch (per cm) (min.)		Breaking Strength pounds (newtons) (min.)		Flex stiffness ($inch\text{-}lbs \times 10^{-4}$) (max) (newton meter)	
Min.	Max.	Warp	Filling	Warp	Filling	Warp	Filling
3.4 (115.26)	3.6 (122.04)	56 (22)	56 (22)	225 (1001)	210 (935)	2.5 (0.283)	1.8 (0.203)

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3.5 Weave The cloth shall be a 3 harness, $\frac{2}{1}$ right hand twill, when tested in accordance with 4.5.

3.6 Width The width shall be as specified (see 6.2), and shall be the minimum width, inclusive of selvage. When woven on shuttleless looms, the width shall be exclusive of selvages.

3.7 Color Unless otherwise specified (see 6.2), the finished cloth shall be shade Green 3410 and shall match the standard sample.

3.7.1 Sulfur content The use of sulfur dyes or dyes containing elementary sulfur or sulfur compounds capable of oxidation to sulfuric acid is prohibited. The finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.5.

*3.7.2 Matching The finishd cloth shall match the standard sample under artificial daylight, having a color temperature of 7000 (+ 500) Kelvin and shall be a good approximation to the standard sample under incandescent lamplight 2850(+ 100) Kevlin.

3.7.3 Colorfastness The finished cloth shall show fastness to laundering, wet-dry cleaning, crocking and light equal to or better than the standard sample. When no standard is available, the cloth shall show a minimum of "good" fastness to laundering, wet-dry cleaning, and light and a Munsell value for crocking not lower than 8.5. Tests shall be performed as specified in 4.5.

3.8 Finish The cloth shall be scoured and shall be heat treated so that the finished cloth shall conform to the requirements of 3.12. The cloth shall not be calendered.

3.9 Nonfibrous material The chloroform extractable material shall not exceed 1.0 percent when tested as specified in 4.5.

3.10 pH The pH value of the water extract of the finished cloth shall be not lower than 5.0 nor higher than 8.5 when tested as specified in 4.5.

3.11 Copper or manganese The finished cloth shall not contain more than 0.003 percent copper or more than 0.0015 percent manganese when tested as specified in 4.5.

3.12 Heat setting The cloth shall be fully heat set at a minimum temperature of 400°F (204°C). The cloth shall show no appreciable change is distortion or puckering, and not more than 2.0 percent dimensional change in either warp or filling when tested as specified in 4.5.

3.13 Face identification The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "face" at each end of the roll.

3.14 Length and put-up Unless otherwise specified (see 5.1.1 and 6.2), the cloth shal be furnished in continous lengths, each not less than 50 yards (45.5m). Each length shall be put-up in full width rolls as specified in PPP-P-1133 with the face inward.

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3.15 Fiber identification Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.16 Workmanship The finished cloth shall conform to the quality estabished by this specification. The demerit points per 100 square yards ($83.6m^2$), when calculated as specified in Section 4, shall not exceed the established point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3)
2. Quality conformance inspection (see 4.4)

4.3 First article inspection When required (see 3.2), the first article of finished cloth shall be visually inspected for color and finished appearance and shall be tested for chemical and physical properties in accordance with the applicable methods specified in 4.5.

4.4 Quality conformance inspection Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 Examination of the end item Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.4.

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4.4.2.1 Yard-by-yard examination Each roll in the sample shall be examined on face side only. When the total yardage in the roll does not exceed 100 yards (91.4m), the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4m) only 100 yards (91.4m) shall be examined. All defects as defined in Section III of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) (0.91m) shall be scored and assigned demerit points as listed in 4.4.2.1.1, except that only those knots which exceed the limits shown on Sears Fabric Defect Scale (see 6.4), "C" for knots, shall be scored. No linear yard (increments of 1 yard (0.91m) on the measuring device of the inspection machine) from any one roll within sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of the total yardage examined exceeds 20 points. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of two or more individual rolls exceed 30 points. If one roll exceeds 30 points per 100 square yards ($83.6m^2$), a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 30 points per 100 square yards ($83.6m^2$). Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample X 3600}}{\text{Contracted width of cloth (inches) X total yards inspected}} = \text{Points per 100 square yards } (83.6m^2)$$

4.4.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches (7.62cm) or less in any dimension - one point
For defects exceeding 3 inches (7.62cm), but not exceeding 6 inches (15.24cm) in any dimension - two points
For defects exceeding 6 inches (15.24cm), but not exceeding 9 inches (22.86cm) in any dimension - three points.
For defects exceeding 9 inches (22.86cm) in any dimension - four points.

The following defects, when present, shall be scored 4 points for each yard (0.91m) in which they occur:

Baggy, ridgy, or wavy cloth
Width less than specified
Characteristics of finish, hand and shade not equal to the standard sample
Non-uniformity of shade (poor dye penetration, mottled, streaky, or cloudy
Holes, cuts, tears, open places

4.4.2.2 Examination for length.

4.4.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards (1.82m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective with respect to length.

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4.4.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

* 4.4.2.3 Examination for shade. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side-to-side, side-to-center, or end-to-end, shall be cause for rejection of the entire lot represented by the sample.

4.4.2.4 Examination for face identification and non-conformance with the Textile Fiber Products Identification Act During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.4.3 Examination of packaging requirements An examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing and marking comply with the Section 5 requirements of this specification.

4.5 Testing of the end item The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table II shall be followed. The physical and chemical values specified in Section 3, except where otherwise indicated, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 2 yards (1.82m), full width of the finished cloth. The lot size shall be expressed in units of one yard (0.91m). The sample size shall be as follows and the lot shall be unacceptable if one or more sample units fail to meet any requirements specified.

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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Table II - Test methods

Characteristic	Requirement Paragraph	Test Method
Weave	3.5	Visual 1/
Yarn:		
Denier	3.3.1	4021 1/ 2/
Filament	3.3.1	Visual 1/ 2/
Luster	3.3.1	Visual 1/ 2/
Weight	3.4	5041
Yarns per inch		
Warp	3.4	5050
Filling	3.4	5050
Breaking Strength	3.4	5100
Flex stiffness (inch-lbs X 10^{-4}) Warp and filling	3.4	5206
pH	3.10	2811
Nonfibrous material	3.9	2611
Sulfur content	3.7.1	2020
Colorfastness to:		
Laundering	3.7.3	5614 4/
Wet-Dry Cleaning	3.7.3	5622
Crocking	3.7.3	5651
Light	3.7.3	5660
Copper or Manganese content	3.11	2053
Heat test	3.12	see 4.5.1 2/ 3/

1/ One determination per sample unit and the results reported as "pass" or "fail".
2/ A certificate of compliance will be acceptable for this requirement.
3/ One determination per sample unit and the dimensional change reported to the nearest 0.1 percent.
4/ Only the stain on the nylon fibers of the color transfer cloth shall be evaluated.

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4.5.1 Heat test The test specimen shall consist of a square of cloth at least 12 by 12 inches (30.48 by 30.48cm). It shall be laid flat without tension, and marked off with a square 10 by 10 inches (25.4 by 25.4cm), having sides parallel to the warp and filling directions of the fabric. The test specimen shall be placed in an oven maintained at 280°F ± 2° (138°C ± 1°) for a period of two hours. At the end of this period, the specimen shall be removed from the oven, and conditioned under standard atmospheric conditions. The specimen shall then be measured for changes in dimension, and shall be visually compared with the original unheated fabric for distortion, puckering, or change in color.

5. PACKAGING

5.1 Preservation-packaging Packaging shall be level A or C (see 6.2).

5.1.1 Levels A and C The cloth, put up as specified, (see 3.14) shall be packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing Packing shall be level A, B, or C (see 6.2).

5.2.1 Levels A, B, and C The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1133.

6. NOTES

6.1 Intended use The cloth covered by this specification is intended for use as a base material for synthetic rubber coating.

*6.2 Ordering data Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Color of cloth required (see 3.7) if other than specified.

c. Width of cloth required (see 6.3).

d. When first article is required, (see 3.2) the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, quantity, and testing and approval of the first article.

e. Minimum length, if other than specified (see 3.14).

f. Levels of packing and packing requirements (see 5.1 and 5.2).

6.3 Standard sample For access to standard sample, address the procuring activity issuing the invitation for bids (see 3.1).

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*6.4 Fabric defect scales Fabric Defect Replica Kits are available from Sears, Roebuck and Company, Department 817 (Attn: BS623-39), Sears Tower, Chicago, IL 60684.

*6.5 Recycled material It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

6.6 Changes from previous issue The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notation and relationshp to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-19256F	CLOTH, TWILL, NYLON (LOW COUNT, 3.5 OUNCE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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